PROBLEM 51

Particle inside a Hollow Sphere in a Rotating Frame

Original statement · Problems.pdf, p. 36

Non-Inertial Frame + Surface Constraint

A heavy point particle P of mass m is constrained to move inside the surface of a hollow sphere of radius R and center C located at $\mathbf{r}_1^C = R(\mathbf{j}_1 + \mathbf{k}_1)$. The surface of the sphere is smooth. The reference frame S_1 shown in the figure is a non-inertial reference frame, because it rotates with a known angular velocity $\boldsymbol{\omega}_{10} = \Omega(t)\,\mathbf{k}_1$ with respect to inertial space. We introduce the two coordiantes [sic] θ and ϕ as shown in the figure, in order to characterize the configuration of P.

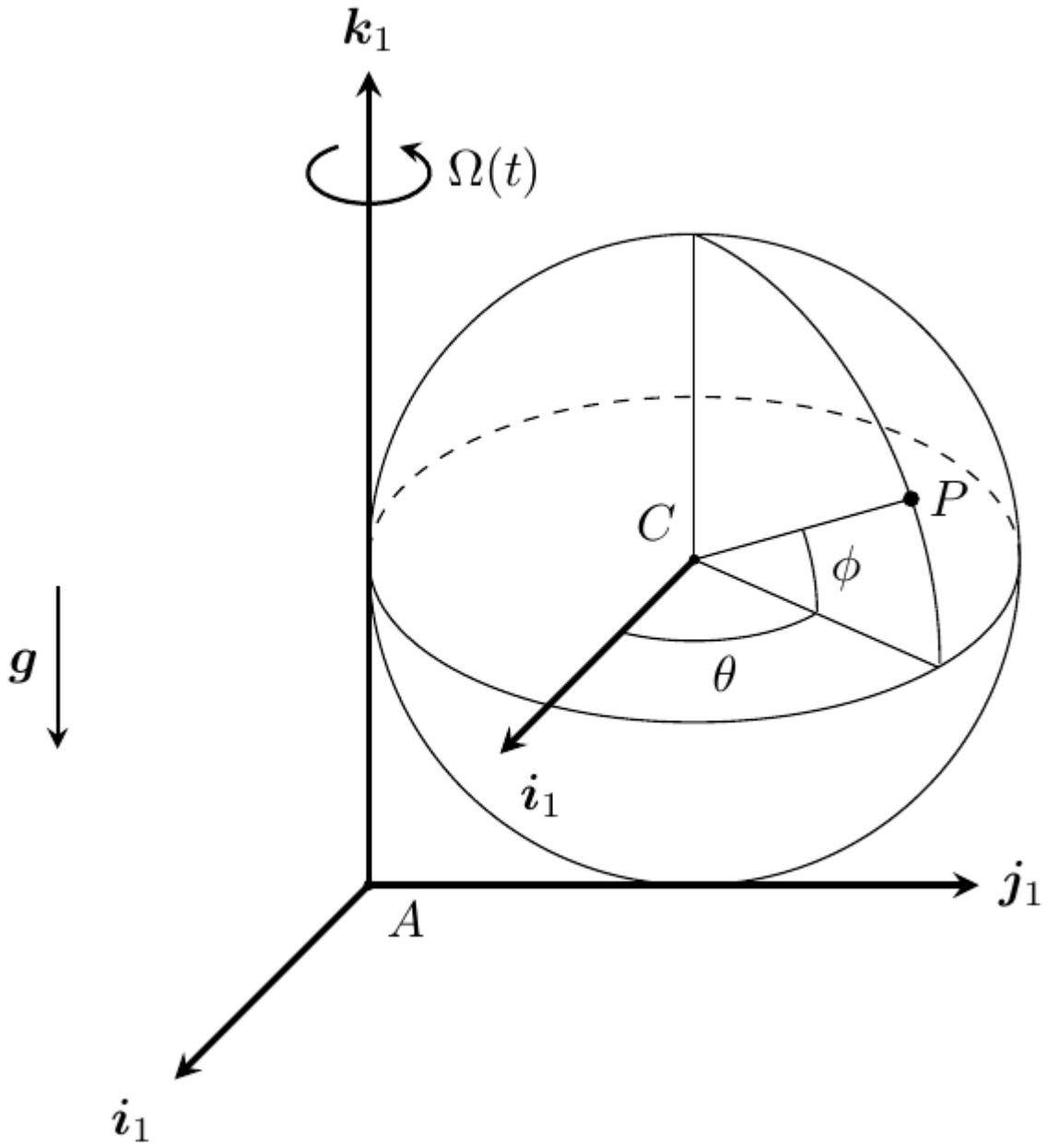


Figure: Rotating frame S1 with axes i1, j1, k1, hollow sphere of centre C at R(j1+k1), particle P located by the angles theta and phi, angular velocity Omega(t) about k1 (Problems.pdf, p. 36).

(a) Give the expressions for the position, velocity, and acceleration vectors of P with respect to S_1, as functions of θ, ϕ and their derivatives. Remember that you may use any unit vectors in your solution, provided that you define them first.

(b) Identify and write down all forces (actual and fictitious) on P as seen from S_1.

(c) Write down two differential equations for θ and ϕ, free of any other unknowns.

(d) Write down an expression for the reaction unknown. This expression may depend on $\theta, \dot{\theta}, \phi, \dot{\phi}$ and the known parameters of the problem. What is the mathematical condition for the particle to separate from the surface?

(e) In the case that $\Omega = \text{const}$, determine and discuss the value of θ, ϕ and the reaction unknown for the equilibrium configurations of P.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a heavy particle P (mass m) slides on the smooth inner surface of a sphere of radius R and centre C, $\mathbf{r}_1^C = R(\mathbf{j}_1 + \mathbf{k}_1)$. The sphere is fixed in the frame $S_1 = \{A; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, which turns about its vertical axis $A\mathbf{k}_1$ with the known rate $\Omega(t)$. The sphere touches the rotation axis at the point $(0, 0, R)$ and the plane $z_1 = 0$ at $(0, R, 0)$.
- **Why S_1.** The constraint (the sphere) is fixed in S_1, so in S_1 the particle simply moves on a fixed sphere described by two angles; all the effect of the rotation is collected in the inertia forces (Notes Eqs. 4.62–4.63). In S_0 the sphere would be a moving constraint.
- **Hypotheses:**
  1. The point A is fixed in an inertial frame S_0 (it lies on the rotation axis), so $\mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$; $\mathbf{k}_1$ is the ascending vertical and gravity is $-g\,\mathbf{k}_1$ (figure).
  2. The surface is smooth: the reaction is normal to the sphere, along the radius CP. It is unilateral: the particle is inside, so the surface can only push it toward C.
  3. $\Omega(t)$ is a known function; $\dot{\Omega}(t)$ is then known too.
- **Frames.** $S_0 = \{A; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, with $\mathbf{k}_0 = \mathbf{k}_1$; S_1, basis $\mathcal{B}_1$, turns about $\mathbf{k}_1$ through the angle $\Psi(t)$, with $\dot{\Psi} = \Omega(t)$.
- **Coordinates (from the figure):** θ is the azimuth about the vertical through C, measured from the direction $\mathbf{i}_1$ toward $\mathbf{j}_1$; ϕ is the elevation of CP above the horizontal plane through C. Ranges: $\theta \in (-\pi, \pi]$, $\phi \in [-\pi/2, \pi/2]$ (the poles $\phi = \pm\pi/2$ are coordinate singularities).
- **Spherical basis at P:** $\{\mathbf{u}_r, \mathbf{u}_\theta, \mathbf{u}_\phi\}$, deduced in the next section.
- **Degrees of freedom:** 3 coordinates minus 1 constraint $|CP| = R$: **2 DOF**, (θ, ϕ). The reaction adds **1 unknown**, N.

Symbol	Meaning	SI unit
m	mass of P	kg
R	radius of the sphere	m
g	gravitational acceleration	m/s^2
$\Omega(t),\ \dot{\Omega}(t)$	angular velocity and acceleration of S_1 about $\mathbf{k}_1$	rad/s, rad/s^2
$\theta,\ \phi$	azimuth and elevation of P seen from C	rad
N	magnitude of the reaction, $\mathbf{N} = -N\,\mathbf{u}_r$, $N \geq 0$ while in contact	N

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

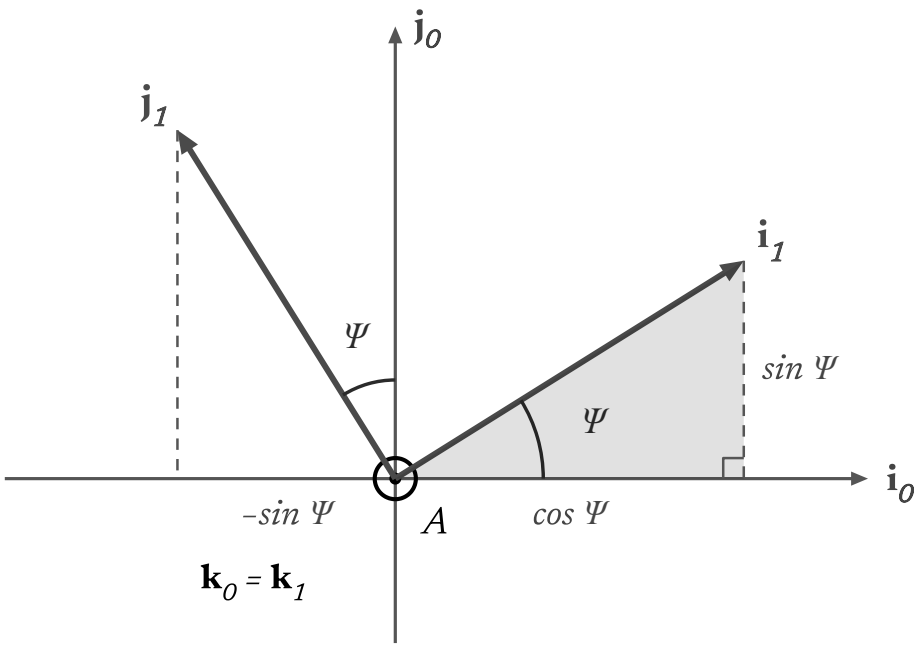


Figure 1. View from $+z$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$). S_1 has turned the angle $\Psi(t)$, $\dot{\Psi} = \Omega(t)$, counter-clockwise: $\mathbf{i}_1$ has the legs $\cos\Psi$ along $\mathbf{i}_0$ and $\sin\Psi$ along $\mathbf{j}_0$; the tip of $\mathbf{j}_1$ moves to the side of $-\mathbf{i}_0$, hence $-\sin\Psi$.

The rotation is about $\mathbf{k}_1 = \mathbf{k}_0$, so we look down from $+z$: $\mathbf{k}_0$ points out of the paper ($\odot$), $\mathbf{i}_0$ to the right and, since $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$ needs a counter-clockwise turn from $\mathbf{i}_0$ to $\mathbf{j}_0$ for the thumb to point at the reader, $\mathbf{j}_0$ points up. A positive Ω turns S_1 counter-clockwise by $\Psi(t)$:

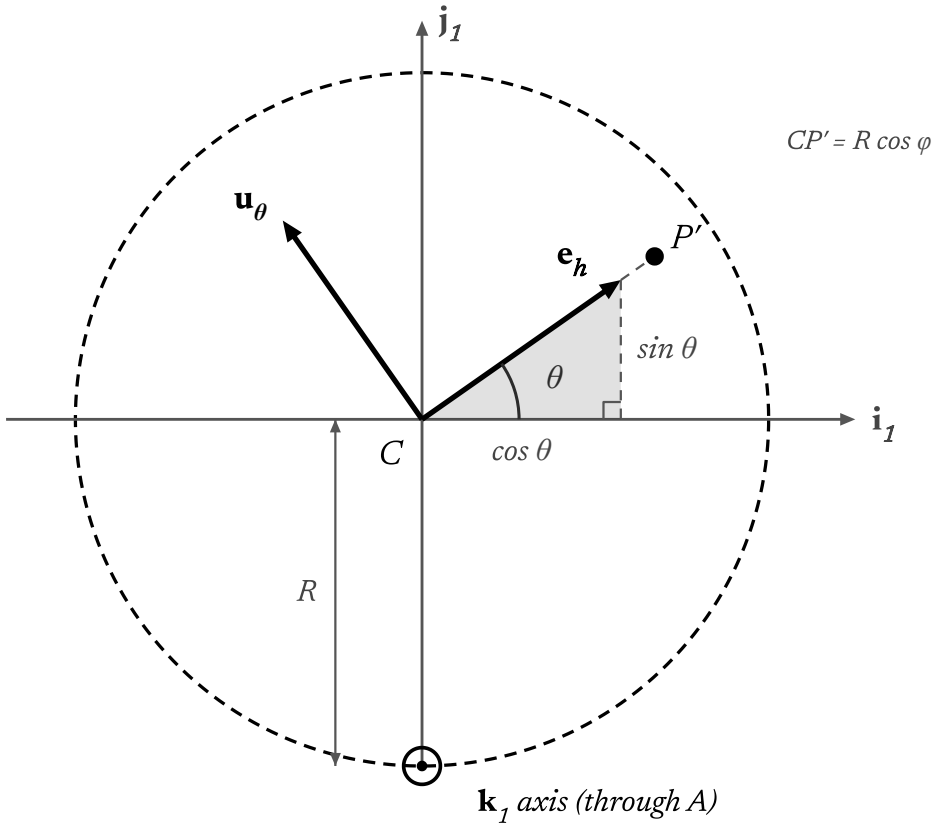
- $\mathbf{i}_1$ is the hypotenuse (length 1) of the shaded triangle: adjacent leg $\cos\Psi$ along $\mathbf{i}_0$, opposite leg $\sin\Psi$ along $+\mathbf{j}_0$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ and its tip moves to the side of $-\mathbf{i}_0$: leg $\cos\Psi$ along $\mathbf{j}_0$ and $\sin\Psi$ towards $-\mathbf{i}_0$, **hence the minus sign** in $\mathbf{j}_1 = -\sin\Psi\,\mathbf{i}_0 + \cos\Psi\,\mathbf{j}_0$.
- $\mathbf{k}_1 = \mathbf{k}_0$.

$$[{}_0R_1] = \begin{pmatrix} \cos\Psi & -\sin\Psi & 0 \\ \sin\Psi & \cos\Psi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1\cdot(\cos\Psi\cos\Psi - (-\sin\Psi)\sin\Psi) = \cos^2\Psi + \sin^2\Psi = 1$$

The columns have unit norm and the first two are orthogonal ($-\cos\Psi\sin\Psi + \sin\Psi\cos\Psi = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$; since $\mathbf{k}_1$ is fixed in both frames, $\boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot{\Omega}\,\mathbf{k}_1 + \Omega\,\boldsymbol{\omega}_{10} \times \mathbf{k}_1 = \dot{\Omega}\,\mathbf{k}_1$; $\mathbf{r}_0^A = \mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$. Check: by the chain rule, $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = \dot{\Psi}\,(-\sin\Psi\,\mathbf{i}_0 + \cos\Psi\,\mathbf{j}_0) = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{i}_1$.

The azimuth θ: horizontal plane through C



dashed circle: equator of the sphere (radius R)

Figure 2. Horizontal plane through C seen from above ($\mathbf{k}_1$ out of the paper, $\odot$). The azimuth θ is measured at C from $\mathbf{i}_1$ towards $\mathbf{j}_1$: $\mathbf{e}_h = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1$ (legs of the shaded triangle) and $\mathbf{u}_\theta = d\mathbf{e}_h/d\theta$. P projects to P' at distance $R\cos\phi$ from C; the axis of rotation is at distance R from C, on the side of $-\mathbf{j}_1$.

Seen from above ($\mathbf{k}_1$ out of the paper), with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up, the centre C is at distance R from the rotation axis on the side of $+\mathbf{j}_1$ (its horizontal position is $R\,\mathbf{j}_1$). The azimuth θ is measured at C from $\mathbf{i}_1$ towards $\mathbf{j}_1$, counter-clockwise in this view. The horizontal unit vector in that direction is the hypotenuse of the shaded triangle, with the leg $\cos\theta$ along $\mathbf{i}_1$ (adjacent) and $\sin\theta$ along $\mathbf{j}_1$ (opposite), and its derivative with respect to θ ($\frac{d}{d\theta}\cos\theta = -\sin\theta$, $\frac{d}{d\theta}\sin\theta = \cos\theta$) is the horizontal unit vector of increasing θ:

$$\mathbf{e}_h = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1, \qquad \mathbf{u}_\theta = \frac{d\mathbf{e}_h}{d\theta} = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$$

(the minus sign: turning $\mathbf{e}_h$ a further 90° counter-clockwise takes its tip to the side of $-\mathbf{i}_1$). Also $\mathbf{k}_1 \times \mathbf{e}_h = \mathbf{u}_\theta$ and $\mathbf{k}_1 \times \mathbf{u}_\theta = -\mathbf{e}_h$.

The elevation ϕ: vertical plane through C and P

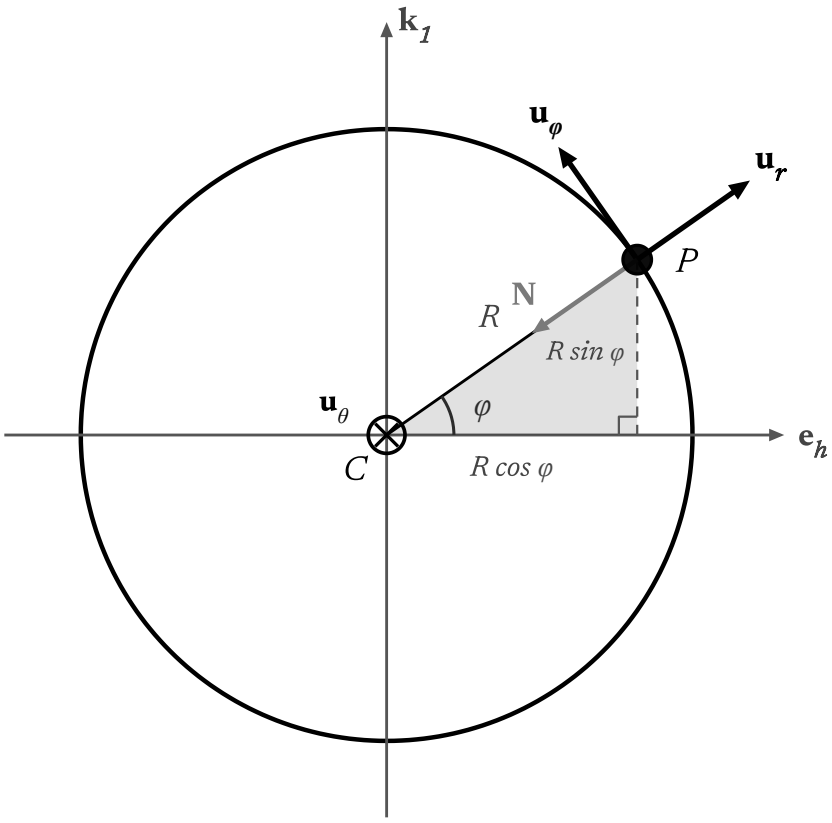


Figure 3. Vertical plane through C that contains $\mathbf{e}_h$ ($\mathbf{u}_\theta$ points into the paper, $\otimes$). The elevation ϕ is measured at C from the horizontal: in the shaded triangle the hypotenuse is R, the leg adjacent to ϕ is $R\cos\phi$ along $\mathbf{e}_h$ and the opposite one $R\sin\phi$ along $\mathbf{k}_1$. $\mathbf{u}_r, \mathbf{u}_\phi$ in orange; the reaction $\mathbf{N} = -N\,\mathbf{u}_r$ (pink) pushes towards C.

P lies in the vertical plane through C that contains $\mathbf{e}_h$; it is drawn with $\mathbf{e}_h$ to the right and $\mathbf{k}_1$ up, so $\mathbf{e}_h \times \mathbf{k}_1 = -\mathbf{u}_\theta$ points out of the paper and $\mathbf{u}_\theta$ points into it ($\otimes$). In the shaded right triangle the hypotenuse is $|CP| = R$, the leg adjacent to ϕ is $R\cos\phi$ along $\mathbf{e}_h$ (the distance CP' of the horizontal view) and the leg opposite to ϕ is $R\sin\phi$ along $\mathbf{k}_1$. The unit vector from C to P and its derivative with respect to ϕ are

$$\mathbf{u}_r = \cos\phi\,\mathbf{e}_h + \sin\phi\,\mathbf{k}_1 = \cos\phi\cos\theta\,\mathbf{i}_1 + \cos\phi\sin\theta\,\mathbf{j}_1 + \sin\phi\,\mathbf{k}_1 \quad (\text{from } C \text{ to } P)$$

$$\mathbf{u}_\phi = \frac{\partial \mathbf{u}_r}{\partial \phi} = -\sin\phi\,\mathbf{e}_h + \cos\phi\,\mathbf{k}_1 = -\sin\phi\cos\theta\,\mathbf{i}_1 - \sin\phi\sin\theta\,\mathbf{j}_1 + \cos\phi\,\mathbf{k}_1 \quad (\text{increasing } \phi)$$

while $\frac{\partial \mathbf{u}_r}{\partial \theta} = \cos\phi\,\frac{d\mathbf{e}_h}{d\theta} = \cos\phi\,\mathbf{u}_\theta$, so $\mathbf{u}_\theta = \frac{1}{\cos\phi}\frac{\partial \mathbf{u}_r}{\partial \theta}$ (horizontal, increasing θ).

Relation with the standard spherical coordinates. The usual polar angle ϑ is measured from the upward vertical $\mathbf{k}_1$ (colatitude), with $\mathbf{u}_r = \sin\vartheta\cos\theta\,\mathbf{i}_1 + \sin\vartheta\sin\theta\,\mathbf{j}_1 + \cos\vartheta\,\mathbf{k}_1$. The elevation is measured from the horizontal instead, so $\phi = \frac{\pi}{2} - \vartheta$, and

$$\sin\vartheta = \sin\left(\tfrac{\pi}{2} - \phi\right) = \sin\tfrac{\pi}{2}\cos\phi - \cos\tfrac{\pi}{2}\sin\phi = \cos\phi, \qquad \cos\vartheta = \cos\left(\tfrac{\pi}{2} - \phi\right) = \cos\tfrac{\pi}{2}\cos\phi + \sin\tfrac{\pi}{2}\sin\phi = \sin\phi$$

which turns the standard $\mathbf{u}_r$ into ours. Because ϕ grows upwards while ϑ grows downwards, $\mathbf{u}_\phi = -\mathbf{u}_\vartheta$.

Change-of-basis matrix (columns: $\mathbf{u}_r, \mathbf{u}_\theta, \mathbf{u}_\phi$ in $\mathcal{B}_1$):

$$[{}_1R_u] = \begin{pmatrix} \cos\phi\cos\theta & -\sin\theta & -\sin\phi\cos\theta \\ \cos\phi\sin\theta & \cos\theta & -\sin\phi\sin\theta \\ \sin\phi & 0 & \cos\phi \end{pmatrix}$$

Checks: each column has unit norm ($\cos^2\phi(\cos^2\theta + \sin^2\theta) + \sin^2\phi = 1$, etc.); $\mathbf{u}_r \cdot \mathbf{u}_\theta = -\cos\phi\cos\theta\sin\theta + \cos\phi\sin\theta\cos\theta = 0$, $\mathbf{u}_r \cdot \mathbf{u}_\phi = -\cos\phi\sin\phi + \sin\phi\cos\phi = 0$, $\mathbf{u}_\theta \cdot \mathbf{u}_\phi = \sin\phi\sin\theta\cos\theta - \sin\phi\sin\theta\cos\theta = 0$, so $[{}_1R_u]^T[{}_1R_u] = I$. The determinant is the triple product $\det[{}_1R_u] = \mathbf{u}_r \cdot (\mathbf{u}_\theta \times \mathbf{u}_\phi)$, and $\mathbf{u}_\theta \times \mathbf{u}_\phi = -\sin\phi\,(\mathbf{u}_\theta \times \mathbf{e}_h) + \cos\phi\,(\mathbf{u}_\theta \times \mathbf{k}_1) = \sin\phi\,\mathbf{k}_1 + \cos\phi\,\mathbf{e}_h = \mathbf{u}_r$ (using $\mathbf{u}_\theta \times \mathbf{e}_h = -\mathbf{k}_1$ and $\mathbf{u}_\theta \times \mathbf{k}_1 = \mathbf{e}_h$), hence $\det[{}_1R_u] = \mathbf{u}_r \cdot \mathbf{u}_r = 1$: the triad $\{\mathbf{u}_r, \mathbf{u}_\theta, \mathbf{u}_\phi\}$ is right-handed.

Useful inverse relations (read from the two triangles): $\mathbf{e}_h = \cos\phi\,\mathbf{u}_r - \sin\phi\,\mathbf{u}_\phi$, $\mathbf{k}_1 = \sin\phi\,\mathbf{u}_r + \cos\phi\,\mathbf{u}_\phi$, $\mathbf{i}_1 = \cos\theta\,\mathbf{e}_h - \sin\theta\,\mathbf{u}_\theta$, $\mathbf{j}_1 = \sin\theta\,\mathbf{e}_h + \cos\theta\,\mathbf{u}_\theta$; and $\mathbf{k}_1 \times \mathbf{e}_h = \mathbf{u}_\theta$, $\mathbf{k}_1 \times \mathbf{u}_\theta = -\mathbf{e}_h$, $\mathbf{k}_1 \times \mathbf{u}_\phi = -\sin\phi\,\mathbf{u}_\theta$.

Derivatives in S_1 (chain rule, θ and ϕ depend on t):

$$\left.\frac{d\mathbf{u}_r}{dt}\right|_1 = \frac{\partial \mathbf{u}_r}{\partial \theta}\dot{\theta} + \frac{\partial \mathbf{u}_r}{\partial \phi}\dot{\phi} = \dot{\theta}\cos\phi\,\mathbf{u}_\theta + \dot{\phi}\,\mathbf{u}_\phi$$

$$\left.\frac{d\mathbf{u}_\theta}{dt}\right|_1 = -\dot{\theta}\,\mathbf{e}_h = -\dot{\theta}\left(\cos\phi\,\mathbf{u}_r - \sin\phi\,\mathbf{u}_\phi\right), \qquad \left.\frac{d\mathbf{u}_\phi}{dt}\right|_1 = -\dot{\phi}\cos\phi\,\mathbf{e}_h - \sin\phi\,\dot{\theta}\,\mathbf{u}_\theta - \dot{\phi}\sin\phi\,\mathbf{k}_1 = -\dot{\phi}\,\mathbf{u}_r - \dot{\theta}\sin\phi\,\mathbf{u}_\theta$$

Position vector

To reach P from A we pass through the centre C, because the distance from C is the one that stays constant:

$$\mathbf{r}_1^P = \overrightarrow{AC} + \overrightarrow{CP} = R\left(\mathbf{j}_1 + \mathbf{k}_1\right) + R\,\mathbf{u}_r$$

In components of $\mathcal{B}_1$: $x_1 = R\cos\phi\cos\theta$, $y_1 = R(1+\cos\phi\sin\theta)$, $z_1 = R(1+\sin\phi)$.

Kinematics

(a) Position, velocity and acceleration of P relative to S_1

Pedagogical Justification: relative quantities are derivatives in S_1; $\mathbf{r}_1^C$ is constant in S_1 and only $\mathbf{u}_r$ changes, through $\theta(t)$ and $\phi(t)$.

$$\boxed{\mathbf{r}_1^P = \mathbf{r}_1^C + \overrightarrow{CP} = R\left(\mathbf{j}_1 + \mathbf{k}_1\right) + R\,\mathbf{u}_r}$$

$$\boxed{\mathbf{v}_1^P = R\left.\frac{d\mathbf{u}_r}{dt}\right|_1 = R\dot{\theta}\cos\phi\,\mathbf{u}_\theta + R\dot{\phi}\,\mathbf{u}_\phi}$$

Differentiating again in S_1 (product rule, $\frac{d}{dt}(\dot\theta\cos\phi) = \ddot\theta\cos\phi - \dot\theta\dot\phi\sin\phi$, and the basis derivatives of the change-of-basis section):

$$\mathbf{a}_1^P = R\left(\ddot{\theta}\cos\phi - \dot{\theta}\dot{\phi}\sin\phi\right)\mathbf{u}_\theta + R\dot{\theta}\cos\phi\left[-\dot{\theta}\left(\cos\phi\,\mathbf{u}_r - \sin\phi\,\mathbf{u}_\phi\right)\right] + R\ddot{\phi}\,\mathbf{u}_\phi + R\dot{\phi}\left(-\dot{\phi}\,\mathbf{u}_r - \dot{\theta}\sin\phi\,\mathbf{u}_\theta\right)$$

Collecting components:

$$\boxed{\mathbf{a}_1^P = -R\left(\dot{\phi}^2 + \dot{\theta}^2\cos^2\phi\right)\mathbf{u}_r + R\left(\ddot{\theta}\cos\phi - 2\dot{\theta}\dot{\phi}\sin\phi\right)\mathbf{u}_\theta + R\left(\ddot{\phi} + \dot{\theta}^2\sin\phi\cos\phi\right)\mathbf{u}_\phi}$$

Forces and Equations of Motion

(b) Forces on P seen from S_1

Pedagogical Justification: in S_1, Notes Eq. 4.63 reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, where $\mathbf{F}_i$ (Notes Eq. 4.62) has four terms (Slide 28). We evaluate each with $\mathbf{a}_0^A = \mathbf{0}$, $\boldsymbol{\omega}_{10} = \Omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \dot{\Omega}\mathbf{k}_1$, and project on the spherical basis.

Actual forces.

- Weight: $m\mathbf{g} = -mg\,\mathbf{k}_1 = -mg\sin\phi\,\mathbf{u}_r - mg\cos\phi\,\mathbf{u}_\phi$.
- Reaction (smooth, unilateral, toward C, as drawn in the vertical view): $\mathbf{N} = -N\,\mathbf{u}_r$, $N \geq 0$ while in contact.

Fictitious forces. Only the part of $\mathbf{r}_1^P$ normal to $\mathbf{k}_1$ enters the cross products with $\mathbf{k}_1$. That part is $\mathbf{r}_\perp = x_1\mathbf{i}_1 + y_1\mathbf{j}_1$; substituting $\mathbf{i}_1 = \cos\theta\,\mathbf{e}_h - \sin\theta\,\mathbf{u}_\theta$ and $\mathbf{j}_1 = \sin\theta\,\mathbf{e}_h + \cos\theta\,\mathbf{u}_\theta$:

$$\mathbf{r}_\perp = R\left[\cos\phi\cos^2\theta + \sin\theta + \cos\phi\sin^2\theta\right]\mathbf{e}_h + R\left[-\cos\phi\cos\theta\sin\theta + \cos\theta + \cos\phi\sin\theta\cos\theta\right]\mathbf{u}_\theta = R\left(\cos\phi + \sin\theta\right)\mathbf{e}_h + R\cos\theta\,\mathbf{u}_\theta$$

- **Origin force:** $\mathbf{F}_A = -m\mathbf{a}_0^A = \mathbf{0}$.
- **Euler force** (Slide 31), with $\mathbf{k}_1 \times \mathbf{e}_h = \mathbf{u}_\theta$, $\mathbf{k}_1 \times \mathbf{u}_\theta = -\mathbf{e}_h$:

$$\mathbf{F}_E = -m\dot{\Omega}\,\mathbf{k}_1 \times \mathbf{r}_\perp = -m\dot{\Omega}R\left[\left(\cos\phi + \sin\theta\right)\mathbf{u}_\theta - \cos\theta\,\mathbf{e}_h\right]$$

$$\mathbf{F}_E = m\dot{\Omega}R\left[\cos\theta\cos\phi\,\mathbf{u}_r - (\cos\phi+\sin\theta)\,\mathbf{u}_\theta - \cos\theta\sin\phi\,\mathbf{u}_\phi\right]$$

- **Centrifugal force** (Slide 30), $\mathbf{F}_C = m\Omega^2\mathbf{r}_\perp$:

$$\mathbf{F}_C = m\Omega^2 R\left[(\cos\phi+\sin\theta)\cos\phi\,\mathbf{u}_r + \cos\theta\,\mathbf{u}_\theta - (\cos\phi+\sin\theta)\sin\phi\,\mathbf{u}_\phi\right]$$

- **Coriolis force** (Slide 31): $\mathbf{k}_1\times\mathbf{v}_1^P = R\dot\theta\cos\phi\,(-\mathbf{e}_h) + R\dot\phi\,(-\sin\phi\,\mathbf{u}_\theta)$, so

$$\mathbf{F}_{COR} = -2m\Omega\,\mathbf{k}_1\times\mathbf{v}_1^P = 2m\Omega R\left(\dot\theta\cos\phi\,\mathbf{e}_h + \dot\phi\sin\phi\,\mathbf{u}_\theta\right) = 2m\Omega R\left[\dot\theta\cos^2\phi\,\mathbf{u}_r + \dot\phi\sin\phi\,\mathbf{u}_\theta - \dot\theta\sin\phi\cos\phi\,\mathbf{u}_\phi\right]$$

As required, $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 2m\Omega R^2\left(\dot\phi\sin\phi\,\dot\theta\cos\phi - \dot\theta\sin\phi\cos\phi\,\dot\phi\right) = 0$.

(c) Two differential equations for θ and ϕ

> **Pedagogical Justification:** the reaction is along $\mathbf{u}_r$, so the projections of Notes Eq. 4.63 on $\mathbf{u}_\theta$ and $\mathbf{u}_\phi$ contain no unknown force.

Projection on $\mathbf{u}_\theta$ (weight and reaction have no $\mathbf{u}_\theta$ component):

$$mR\left(\ddot\theta\cos\phi - 2\dot\theta\dot\phi\sin\phi\right) = m\Omega^2 R\cos\theta - m\dot\Omega R\,(\cos\phi+\sin\theta) + 2m\Omega R\dot\phi\sin\phi$$

$$\boxed{\ddot\theta\cos\phi - 2\dot\phi\sin\phi\left(\dot\theta+\Omega\right) = \Omega^2\cos\theta - \dot\Omega\,(\cos\phi+\sin\theta)}$$

Projection on $\mathbf{u}_\phi$:

$$mR\left(\ddot\phi + \dot\theta^2\sin\phi\cos\phi\right) = -mg\cos\phi - m\Omega^2 R\,(\cos\phi+\sin\theta)\sin\phi - m\dot\Omega R\cos\theta\sin\phi - 2m\Omega R\dot\theta\sin\phi\cos\phi$$

$$\boxed{\ddot{\phi} + \dot{\theta}\left(\dot{\theta} + 2\Omega\right)\sin\phi\cos\phi + \frac{g}{R}\cos\phi + \Omega^2\sin\phi\left(\cos\phi + \sin\theta\right) + \dot{\Omega}\cos\theta\sin\phi = 0}$$

These two coupled, non-linear ODEs with the known forcing $\Omega(t)$ determine $\theta(t)$, $\phi(t)$ from the initial conditions.

Projection on $\mathbf{u}_r$ (it contains the only unknown force, N):

$$-mR\left(\dot{\phi}^2 + \dot{\theta}^2\cos^2\phi\right) = -N - mg\sin\phi + m\Omega^2 R\cos\phi\left(\cos\phi + \sin\theta\right) + m\dot{\Omega}R\cos\theta\cos\phi + 2m\Omega R\dot{\theta}\cos^2\phi$$

Answers and Checks

(a) Kinematics

$\mathbf{r}_1^P$, $\mathbf{v}_1^P$ and $\mathbf{a}_1^P$ are the three boxed expressions of the kinematics section, with $\mathbf{u}_r$, $\mathbf{u}_\theta$, $\mathbf{u}_\phi$ defined in the change-of-basis section.

(b) Forces

Weight $-mg\,\mathbf{k}_1$, reaction $-N\,\mathbf{u}_r$, origin force $\mathbf{0}$, and the Euler, centrifugal and Coriolis forces written above.

(c) Equations of motion

The two boxed equations for θ and ϕ.

(d) Reaction and separation condition

From the projection on $\mathbf{u}_r$:

$$\boxed{N = mR\left(\dot{\phi}^2 + \dot{\theta}^2\cos^2\phi\right) - mg\sin\phi + m\Omega^2 R\cos\phi\left(\cos\phi + \sin\theta\right) + m\dot{\Omega}R\cos\phi\cos\theta + 2m\Omega R\dot{\theta}\cos^2\phi}$$

Grouping $\dot{\theta}^2 + 2\Omega\dot{\theta} + \Omega^2 = (\dot{\theta} + \Omega)^2$ in the $\cos^2\phi$ terms gives the equivalent compact form

$$N = mR\dot{\phi}^2 + mR\cos^2\phi\left(\dot{\theta} + \Omega\right)^2 + m\Omega^2 R\cos\phi\sin\theta + m\dot{\Omega}R\cos\phi\cos\theta - mg\sin\phi$$

It depends only on $\theta, \dot\theta, \phi, \dot\phi$ and the known $m, R, g, \Omega(t), \dot\Omega(t)$. **Separation:** the surface can only push ($N \geq 0$). The particle leaves the surface at the first instant at which

$$\boxed{N\left(\theta, \dot\theta, \phi, \dot\phi, t\right) = 0 \ \text{ and would become negative}}$$

i.e. the motion predicted by (c) is valid only while $N \geq 0$.

(e) Equilibrium configurations for $\Omega = \text{const}$

> **Pedagogical Justification:** equilibrium in S_1 means θ, ϕ constant: $\dot\theta = \dot\phi = \ddot\theta = \ddot\phi = 0$, with $\dot\Omega = 0$. Then $\mathbf{F}_E = \mathbf{F}_{COR} = \mathbf{0}$ and the equations of (c) become algebraic. We take $\Omega \neq 0$ (the case $\Omega = 0$ is discussed at the end).

From the θ equation: $\Omega^2 \cos\theta = 0 \Rightarrow \cos\theta = 0$, so $\theta = \pi/2$ or $\theta = -\pi/2$: P must lie in the vertical plane $x_1 = 0$ that contains the rotation axis and C (otherwise the centrifugal force has a component $m\Omega^2 R\cos\theta$ along $\mathbf{u}_\theta$ that nothing balances).

From the ϕ equation, multiplied by R: $g\cos\phi + \Omega^2 R\sin\phi(\cos\phi + \sin\theta) = 0$. The poles $\cos\phi = 0$ do not satisfy it ($\Omega^2 R\sin\phi\sin\theta = \pm\Omega^2 R \neq 0$), so we can divide by $g\cos\phi > 0$. With the dimensionless number $\lambda = \Omega^2 R/g > 0$:

Branch 1, $\theta = \pi/2$ (far side of the sphere from the axis, $y_1 = R(1+\cos\phi) \geq R$):

$$1 + \lambda\,\frac{\sin\phi\,(\cos\phi + 1)}{\cos\phi} = 0 \quad\Longleftrightarrow\quad \boxed{\tan\phi_1 + \sin\phi_1 = -\frac{g}{\Omega^2 R}}$$

The function $f(\phi) = \tan\phi + \sin\phi$ is strictly increasing on $(-\pi/2, \pi/2)$ ($f' = \sec^2\phi + \cos\phi > 0$), goes from $-\infty$ to $+\infty$ and vanishes at 0; so there is **exactly one root,** $\phi_1 \in (-\pi/2, 0)$, below the horizontal plane of C.

Branch 2, $\theta = -\pi/2$ (side of the sphere next to the axis, $y_1 = R(1-\cos\phi)$):

$$1 + \lambda\,\frac{\sin\phi\,(\cos\phi - 1)}{\cos\phi} = 0 \quad\Longleftrightarrow\quad \boxed{\tan\phi_2 - \sin\phi_2 = \frac{g}{\Omega^2 R}}$$

$h(\phi) = \tan\phi - \sin\phi$ has $h' = \sec^2\phi - \cos\phi \geq 0$ (zero only at $\phi = 0$), $h(0) = 0$ and $h \to +\infty$ as $\phi \to \pi/2$, so there is **exactly one root,** $\phi_2 \in (0, \pi/2)$, above the horizontal plane of C.

Reaction at equilibrium. With $\dot\theta = \dot\phi = \dot\Omega = 0$, (d) gives $N = m\Omega^2 R\cos\phi(\cos\phi + \sin\theta) - mg\sin\phi$. The equilibrium relation $\Omega^2 R(\cos\phi + \sin\theta) = -g\cos\phi/\sin\phi$ ($\sin\phi \neq 0$ at both roots) turns it into

$$N = -\frac{mg\cos^2\phi}{\sin\phi} - mg\sin\phi = -\frac{mg\left(\cos^2\phi + \sin^2\phi\right)}{\sin\phi} \quad\Longrightarrow\quad \boxed{N = -\frac{mg}{\sin\phi}}$$

This is simply the vertical balance: the centrifugal force is horizontal, so the vertical component of $\mathbf{N} = -N\mathbf{u}_r$, which is $-N\sin\phi$ (the leg $\sin\phi$ of the vertical-view triangle), must cancel the weight.

Discussion.

- **Branch 1** ($\theta = \pi/2$, $\phi_1 < 0$): $N_1 = mg/|\sin\phi_1| > 0$. **This is the only physically possible equilibrium.** The particle sits in the lower half of the sphere, on the side away from the axis, where the inward normal $-\mathbf{u}_r$ points up and toward the axis, balancing the weight and the outward centrifugal force. As Ω grows, ϕ_1 rises toward the equator of the sphere: for $\lambda \gg 1$, $\phi_1 \approx -g/(2\Omega^2 R)$ and $N_1 \approx 2m\Omega^2 R$ (the centrifugal force at distance $2R$ from the axis); for $\Omega \to 0$, $\phi_1 \to -\pi/2$ (bottom) and $N_1 \to mg$. Example ($R = 1$ m, $g = 9.81$ m/s^2): $\Omega = 3$ rad/s gives $\phi_1 = -30.3^\circ$ and $N_1 = 1.98\,mg$.
- **Branch 2** ($\theta = -\pi/2$, $\phi_2 > 0$): $N_2 = -mg/\sin\phi_2 < 0$. It solves the equations of (c), but the reaction would have to **pull** the particle away from C, which the inner surface cannot do: it is **not an admissible equilibrium** for the unilateral contact (it would be one for a particle held on both sides of the surface, or resting on the outside of the sphere). On that side the centrifugal force points toward the inside of the sphere, so it cannot help the surface to hold the particle up.
- **Stability of branch 1** (extension, using the energy in S_1). For constant Ω, A fixed and a smooth sphere fixed in S_1, the energy $E_1^P = T_1^P + V_1$ is conserved (Slides 29–31), with $V_1 = mgz_1 - \frac{1}{2}m\Omega^2(x_1^2 + y_1^2) = mgR(1 + \sin\phi) - \frac{1}{2}m\Omega^2R^2\left(1 + \cos^2\phi + 2\cos\phi\sin\theta\right)$. Its derivatives $\partial V_1/\partial\theta = -m\Omega^2R^2\cos\phi\cos\theta$ and $\partial V_1/\partial\phi = mgR\cos\phi + m\Omega^2R^2\sin\phi(\cos\phi + \sin\theta)$ reproduce the equilibrium equations. At branch 1, $\partial^2V_1/\partial\theta^2 = m\Omega^2R^2\cos\phi_1\sin\theta = m\Omega^2R^2\cos\phi_1 > 0$, $\partial^2V_1/\partial\theta\partial\phi = m\Omega^2R^2\sin\phi_1\cos\theta = 0$ and $\partial^2V_1/\partial\phi^2 = mgR\lambda\cos\phi_1\left(\sec^2\phi_1 + \cos\phi_1\right) > 0$: V_1 has a strict minimum, so the equilibrium is stable (the linearised equations give purely imaginary eigenvalues, see the verification).

- **Case $\Omega = 0$:** the θ equation is satisfied identically and the ϕ equation gives $\cos\phi = 0$: the bottom point $\phi = -\pi/2$ with $N = mg$ (the top point $\phi = \pi/2$ would need $N = -mg < 0$); θ is undefined there (pole of the coordinates).

Checks

- **Role of each inertia force.** The centrifugal force pushes P away from the axis and is responsible for the tilt of the equilibrium; the Euler force (only while Ω changes) drags P against the sense of the angular acceleration, along $-\mathbf{u}_\theta$-like directions; the Coriolis force couples the two angular motions (terms $2\Omega\dot\phi\sin\phi$ and $2\Omega\dot\theta\sin\phi\cos\phi$) and does no work.
- **Absolute reading of N.** In the compact form, $\dot\theta + \Omega$ is the absolute azimuthal rate of the particle about the vertical through C, so $mR\dot\phi^2 + mR\cos^2\phi(\dot\theta+\Omega)^2$ is the centripetal demand of the absolute motion on the sphere.
- **Limit $\Omega \to 0$:** the equations reduce to those of the spherical pendulum, $\ddot\theta\cos\phi - 2\dot\theta\dot\phi\sin\phi = 0$ and $\ddot\phi + \dot\theta^2\sin\phi\cos\phi + (g/R)\cos\phi = 0$, and $N = mR(\dot\phi^2 + \dot\theta^2\cos^2\phi) - mg\sin\phi$.
- **Dimensions:** in (c) every term is s^{-2} ($[\ddot\theta] = [\Omega^2] = [\dot\Omega] = [\Omega\dot\phi] = [g/R] = \mathrm{s}^{-2}$); in (d) every term is $\mathrm{kg\,m\,s}^{-2} = \mathrm{N}$; $\lambda = \Omega^2 R/g$ is dimensionless.
- **Verification (sympy):** with a general $\Psi(t)$, $\dot\Psi = \Omega(t)$, the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition; solving $m\mathbf{a}_0^P = -mg\mathbf{k}_1 - N\mathbf{u}_r$ in S_0 returns exactly the two equations of (c) and the N of (d); for $\Omega = 1, 3, 10$ rad/s ($R = 1$ m) both branches satisfy the equations with $N = -mg/\sin\phi$, branch 1 has $N > 0$ and purely imaginary eigenvalues, branch 2 has $N < 0$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 51 in its solution keys, so the results are verified by independent derivation (the sympy check recorded above).

Source: Problems.pdf (PDF page 36); no official key